

Homework #2

Due September 13th, Tuesday, before class starts

1. The median ζ , of a set of numbers is such that half the values in the set are below ζ , and the other half are above it. For example, the median of the set of values $\{12, 3, 8, 20, 21, 75, 31\}$ is 20. Show that an operator that computes the median of a subimage area, S , is nonlinear. (10pts)
Hints: an operator H is linear if $H[a_1f_1(x, y) + a_2f_2(x, y)] = a_1H[f_1(x, y)] + a_2H[f_2(x, y)]$

Solution:

The median, ζ , of a set of numbers is such that half the values in the set are below ζ and the other half are above it. To answer this question, you just need to show a simple example that the definition of linearity is violated by the median operator.

For example, let $S_1 = \{1, -2, 3\}$, $S_2 = \{4, 5, 6\}$, and $a = b = 1$. In this case H is the median operator.

We then have $H(S_1 + S_2) = \text{median}\{5, 3, 9\} = 5$, where it is understood that $S_1 + S_2$ is the array sum of S_1 and S_2 . Next, we compute $H(S_1) = \text{median}\{1, -2, 3\} = 1$ and $H(S_2) = \text{median}\{4, 5, 6\} = 5$.

Then, because $H(aS_1 + bS_2) \neq aH(S_1) + bH(S_2)$, it follows that the definition of linearity is violated and the median is a nonlinear operator.

2. Image subtraction is used often in industrial applications for detecting missing components in product assembly. The approach is to store a “golden” image that corresponds to a correct assembly; this image is then subtracted from incoming images of the same product. Ideally, the differences would be zero if the new products are assembled correctly. Difference images for products with missing components would be nonzero in the area where they differ from the golden image. What conditions do you think have to be met in practice for this method to work? You need to list at least two major factors that affect the detection performance. (20 pts)

Solution:

Let $g(x, y)$ denote the golden image, and let $f(x, y)$ denote any input image acquired during routine operation of the system. Change detection via subtraction is based on computing the simple difference $d(x, y) = g(x, y) - f(x, y)$. The resulting image, $d(x, y)$, can be used in two fundamental ways for change detection. One way is use pixel-by-pixel analysis. In this case we say that $f(x, y)$ is “close enough” to the golden image if all the pixels in $d(x, y)$ fall within a specified threshold band $[T_{min}, T_{max}]$ where T_{min} is negative and T_{max} is positive. Usually, the same value of threshold is used for both negative and positive differences, so that we have a band $[-T, T]$ in which all pixels of $d(x, y)$ must fall in order for $f(x, y)$ to be declared acceptable. The second major approach is simply to sum all the pixels in $d(x, y)$ and compare the sum against a

threshold Q . Note that the absolute value needs to be used to avoid errors canceling out. This is a much cruder test, so we will concentrate on the first approach.

There are three fundamental factors that need tight control for difference based inspection to work:

- (1) proper registration,
- (2) controlled illumination, and
- (3) noise levels that are low enough so that difference values are not affected appreciably by variations due to noise.

The first condition basically addresses the requirement that comparisons be made between corresponding pixels. Two images can be identical, but if they are displaced with respect to each other, comparing the differences between them makes no sense. Often, special markings are manufactured into the product for mechanical or image-based alignment

Controlled illumination (note that “illumination” is not limited to visible light) obviously is important because changes in illumination can affect dramatically the values in a difference image. One approach used often in conjunction with illumination control is intensity scaling based on actual conditions. For example, the products could have one or more small patches of a tightly controlled color, and the intensity (and perhaps even color) of each pixels in the entire image would be modified based on the actual versus expected intensity and/or color of the patches in the image being processed.

Finally, the noise content of a difference image needs to be low enough so that it does not materially affect comparisons between the golden and input images. Good signal strength goes a long way toward reducing the effects of noise. Another (sometimes complementary) approach is to implement image processing techniques (e.g., image averaging) to reduce noise. Obviously there are a number of variations of the basic theme just described. For example, additional intelligence in the form of tests which are more sophisticated than pixel-by-pixel threshold comparisons can be implemented. A technique used often in this regard is to subdivide the golden image into different regions and perform different (usually more than one) tests in each of the regions, based on expected region content.

3. Suppose that a digital image is subjected to histogram equalization. Show that a second pass of histogram equalization (on the histogram-equalized image- will produce exactly the same result as the first pass? (20 pts)

Solution:

Let $n = MN$ be the total number of pixels and let n_{r_j} be the number of pixels in the input image with intensity value r_j . Then, the histogram equalization transformation is

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k \frac{n_{r_j}}{n} = \frac{L - 1}{n} \sum_{j=0}^k n_{r_j}$$

Because every pixel (and no others) with value r_k is mapped to value s_k , it follows that $n_{s_k} = n_{r_k}$. A second pass of histogram equalization would produce values v_k according to the transformation

$$v_k = T(s_k) = \frac{L - 1}{n} \sum_{j=0}^k n_{s_j}$$

But, $n_{s_k} = n_{r_k}$, so

$$v_k = T(s_k) = \frac{L-1}{n} \sum_{j=0}^k n_{r_j} = s_k$$

which shows that a second pass of histogram equalization would yield the same result as the first pass. Note that we have assumed negligible round-off errors.

4. Suppose that a 4-bit image (L=16) of size 64*64 has the intensity distribution as Table A. It is desired to transform the original histogram to a specified histogram as shown in Table B. What will be the actual histogram after the histogram matching operation? (30 pts)

r_k	n_k	z_q	n_q
0	16	0	64
1	0	1	128
2	512	2	512
3	32	3	512
4	64	4	1024
5	128	5	1024
6	64	6	512
7	1024	7	128
8	512	8	128
9	512	9	64
10	16	10	0
11	1024	11	0
12	0	12	0
13	128	13	0
14	0	14	0
15	64	15	0

Table A

Table B

First, we apply histogram equalization on the input image.

$$s_0 = T(r_0) = 15 \sum_{j=0}^0 p_r(r_j) = \frac{15}{64 * 64} * 16 \approx 0.06 \approx 0$$

$$s_1 = T(r_1) = 15 \sum_{j=0}^1 p_r(r_j) \approx 0.06 \approx 0$$

$$s_2 = T(r_2) = 15 \sum_{j=0}^2 p_r(r_j) \approx 1.93 \approx 2$$

Similarly,

$$\begin{aligned} s_3 &\approx 2.05 \approx 2 \\ s_4 &\approx 2.28 \approx 2 \\ s_5 &\approx 2.75 \approx 3 \\ s_6 &\approx 2.99 \approx 3 \\ s_7 &\approx 6.73 \approx 7 \\ s_8 &\approx 8.61 \approx 9 \\ s_9 &\approx 10.49 \approx 10 \\ s_{10} &\approx 10.54 \approx 11 \\ s_{11} &\approx 14.3 \approx 14 \\ s_{12} &\approx 14.3 \approx 14 \\ s_{13} &\approx 14.77 \approx 15 \\ s_{14} &\approx 14.77 \approx 15 \\ s_{15} &= 15 \end{aligned}$$

Second, we apply histogram equalization on the output image using the desired histogram.

$$G(z_0) = 15 \sum_{j=0}^0 p_z(z_j) = \frac{64}{64 * 64} * 16 \approx 0.23 \approx 0$$

Similarly,

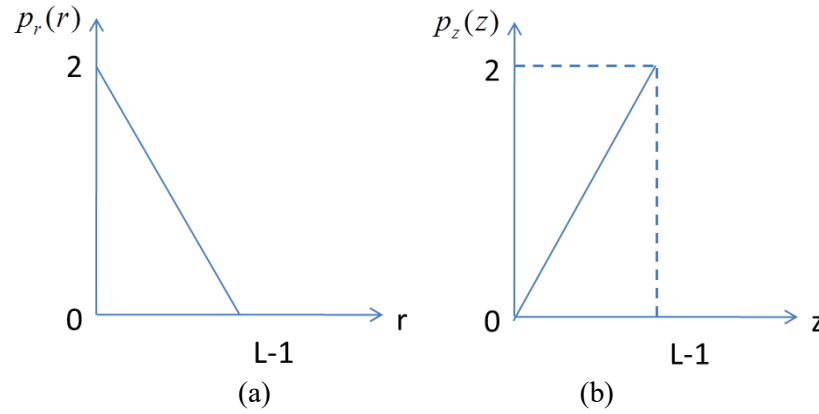
$G(z_1) \approx 0.70 \approx 1$, $G(z_2) \approx 2.58 \approx 3$, $G(z_3) \approx 4.45 \approx 4$, $G(z_4) \approx 8.2 \approx 8$,

$G(z_5) \approx 11.95 \approx 12$, $G(z_6) \approx 13.82 \approx 14$, $G(z_7) \approx 14.29 \approx 14$, $G(z_8) \approx 14.77 \approx 15$

$G(z_9)$ to $G(z_{15}) = 15$

r_k	n_k		s_k		$G(z_q)$		z_q	n_q Actual
0	16	→	0	↗	0	→	0	16
1	0	→	0	↗	1	→	1	608
2	512	→	2	↗	3	→	2	192
3	32	→	2	↗	4	→	3	0
4	64	→	2	↗	8	→	4	2048
5	128	→	3	↗	12	→	5	16
6	64	→	3	↗	14	→	6	1024
7	1024	→	7	↗	14	→	7	0
8	512	→	9	↗	15	→	8	192
9	512	→	10	↗	15	→	9	0
10	16	→	11	↗	15	→	10	0
11	1024	→	14	↗	15	→	11	0
12	0	→	14	↗	15	→	12	0
13	128	→	15	↗	15	→	13	0
14	0	→	15	↗	15	→	14	0
15	64	→	15	↗	15	→	15	0

5. An image with intensities in the range $[0, L-1]$ has the pdf $p_r(r)$ shown in Figure (a). It is desired to transform the intensity levels of this image so that they will have the specified pdf $p_z(z)$ as shown in Figure (b). Assume continuous quantities and find the transformation $z = f(r)$ that will accomplish this. (20 pts)



Solution:

$$s = T(r) = (L-1) \int_0^r p_r(w) dw = (L-1) \int_0^r \left(-\frac{2w}{L-1} + 2 \right) dw = (L-1)2r - r^2 \quad (1)$$

$$s = G(z) = (L-1) \int_0^z p_z(w) dw = (L-1) \int_0^z \frac{2w}{L-1} dw = z^2$$

$$\Rightarrow z = G^{-1}(z) = \pm\sqrt{s}$$

Since the intensity value is non-negative, $z = \sqrt{s}$ (2)

Substitute (1) into (2), we get

$$z = \sqrt{s} = \sqrt{(L-1)2r - r^2}$$